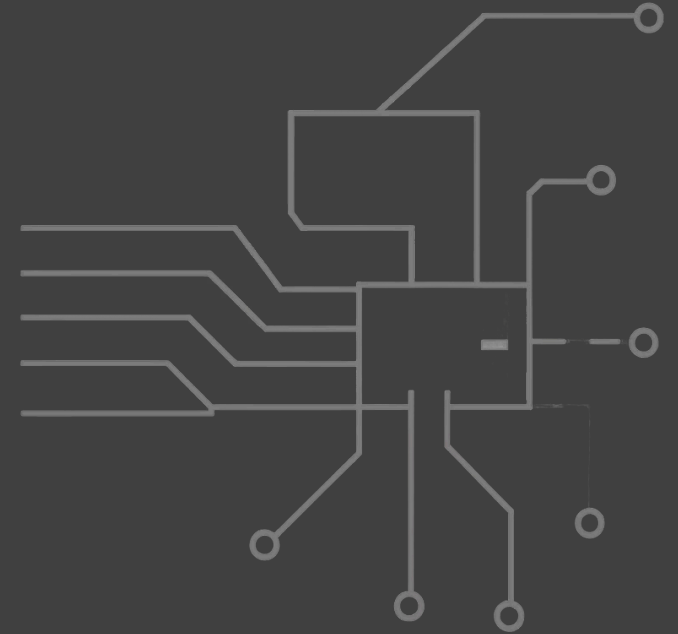


Asynchronous Dataflow

The Evolution of High-Performance Systems

Evelyn Villarreal



Beyond Control-Driven Models

Traditional Execution

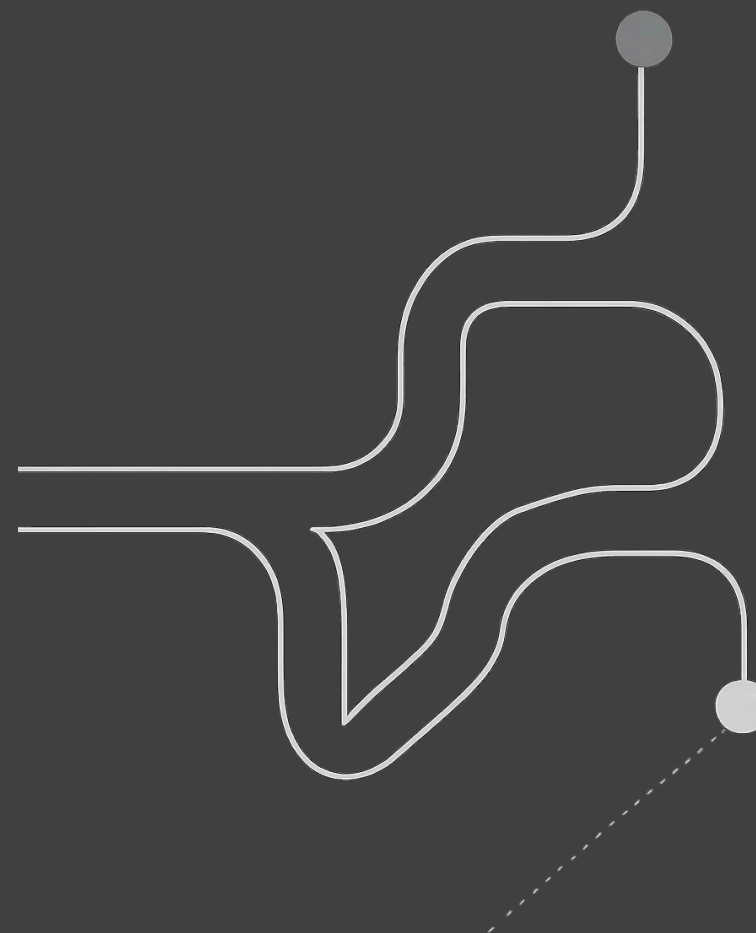
Thread blocking & sequential bottlenecks.

The Dataflow Paradigm

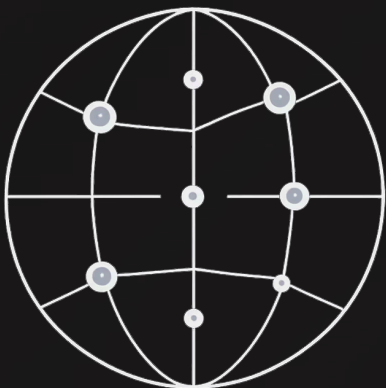
Execution triggers strictly upon data availability.

Impact

Unlocks massive parallelism without arbitrary ordering.



The Architectural Shift



Legacy Bottleneck

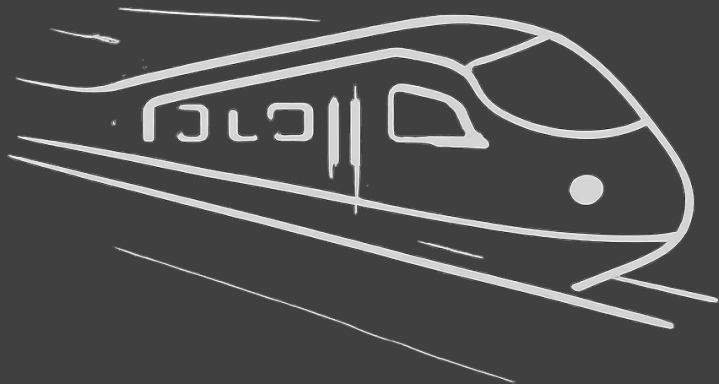
Synchronous barriers & batch processing overhead (e.g., legacy Spark/MapReduce).

The Breakthrough

Decoupling **control-plane latency** from data-plane execution

Core Innovation

Asynchronous operator dispatch driven by data arrival.



Naiad & the Timely Dataflow Revolution

1

Hybrid Execution

Unifying fine-grained **sync + async** processing.

2

Progress Tracking

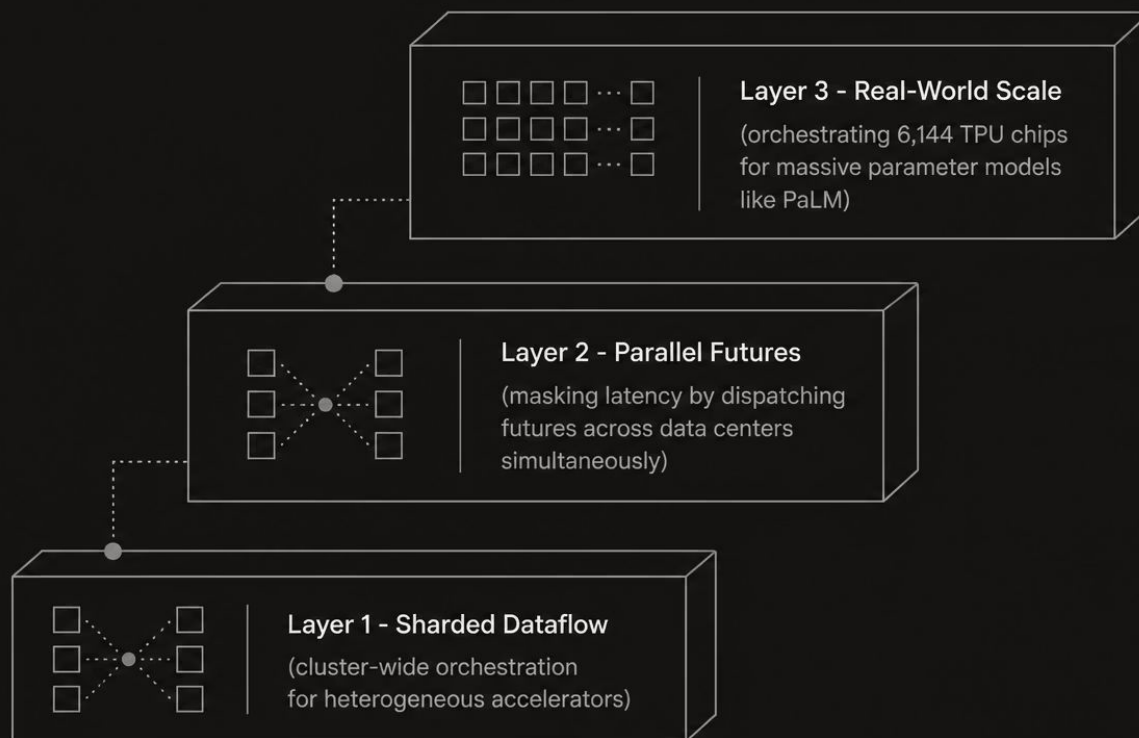
Structured timestamps manage **cyclic dataflow** — no deadlock

3

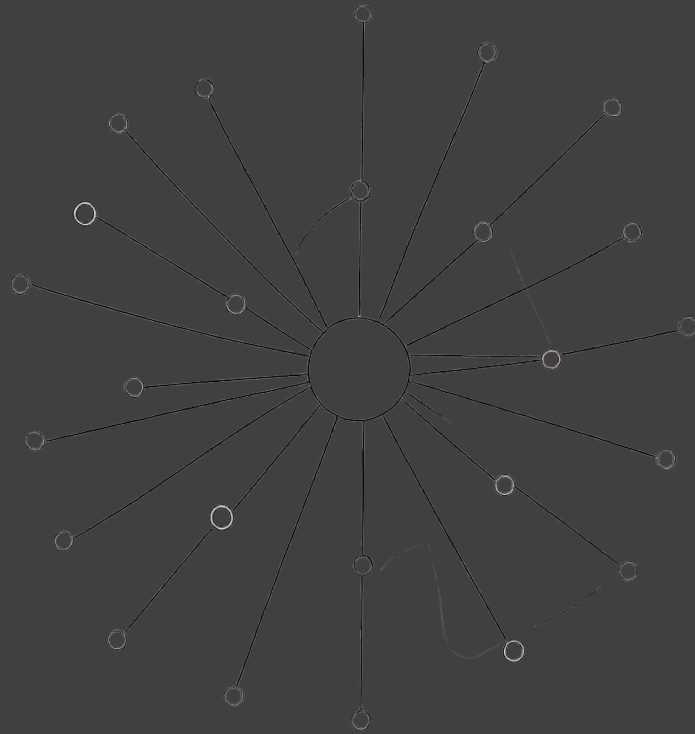
Performance Leap

Sub-millisecond iterations bypassing legacy batch processing

Modern Orchestration: Pathways



TPU (Tensor Processing Unit) is a specialized AI accelerator chip developed by Google to speed up machine learning tasks.



The Future of Data-Driven Systems

Convergence

Timeliness + Consistency +
Expressivity

Next Era

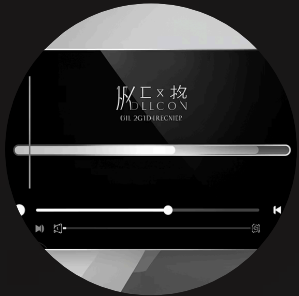
Virtualized async dataflow
powering **global-scale AI**

Core Takeaway

Align architecture with **data's natural flow** — unlock unprecedented scale

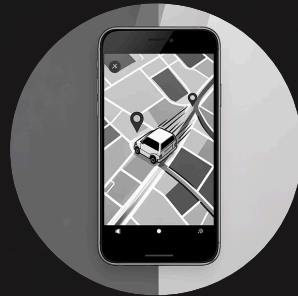
Real-World Web Applications

Asynchronous dataflow principles are already powering many of the web applications we interact with daily, enabling real-time, dynamic experiences.



Netflix & YouTube

Adaptive video streaming
chunk-by-chunk over HTTP for
seamless playback.



Uber & Delivery Apps

Continuous coordinate
updates via WebSocket data
streams for live tracking.



Figma & Collaborative Tools

Real-time, conflict-free state
synchronization across
multiple users.



Financial Dashboards

Low-latency trading and live
ticker updates without page
reloads.

Application Case Study

Real-Time IoT Sensor Data Processing

1

The Software Bridge

Bringing the "Dataflow Rule" to the application layer, ensuring execution triggers only upon data availability.

2

Tech Stack

TypeScript, **Node.js**, and **Serverless PostgreSQL** (Neon/Supabase) provide a modern, scalable foundation.

3

Core Mechanics

Leveraging **Async Generators** (`async function*`) & **Async Iterators** (`for await...of`) for efficient, non-blocking data consumption.